

SCET202

MACHINES & TRANSFORMERS

MODULE:1

Module:3 – MMF, EMF, Torque

Knowledge about constructional details of DC machines

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Module – 1

of dc machines - armature winding- single layer winding, double wave, equalizer rings, dummy coils, **MMF of a winding**, **EMF** **electric torque** - numerical problems.

MMF of windings

Motive force (MMF) is the force which establishes flux in a circuit.

When an electric current is passed through a coil, a magnetic flux is set up in the coil. The MMF of the coil depends on the value of current (I) and number of turns (N) of the coil. It is denoted by MMF and is measured in ampere-turns (AT).

$$\text{MMF} = NI$$

EMF Developed

length 'l' is placed in a magnetic field 'B' and it moves with velocity 'v'.

It moves a distance 'dx' in time 'dt' as shown in the diagram.

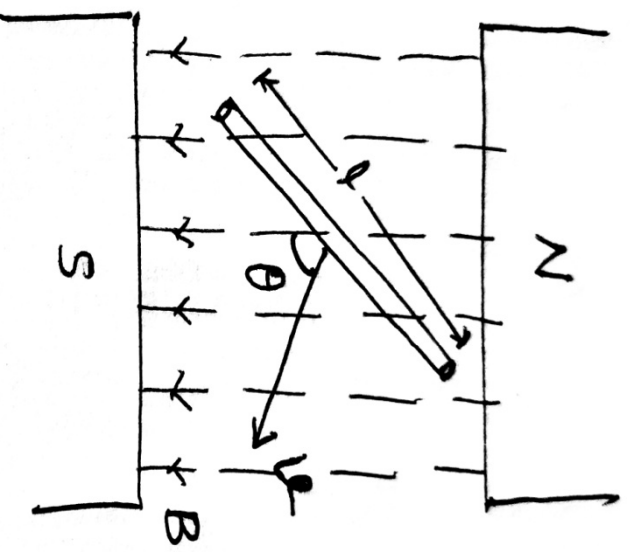
Change in flux = $l \times dx \times B$

According to Faraday's law, induced emf will be equal to the rate of change of flux.

$$\frac{d\phi}{dt} = \frac{l \times dx \times B}{dt} = Blv$$

When the velocity and field is 'v' and 'B' respectively, the induced emf is 'e'.

$$e = Blv \sin \theta$$



Magnetic Torque Developed

of conductor of length
magnetic field 'B' and
current 'I'

experienced in the

$$B l \sin \theta$$

angle between current

$$= B l \sin \theta \times R$$

